

Under review as a conference paper at ICAIS 2025

000 ENHANCING IMAGE GENERATION WITH MULTI-
001 MODAL VQ-VAE AND SELF-SUPERVISED LEARNING
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005 Anonymous authors
006 Paper under double-blind review
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008
009 ABSTRACT
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011 This paper addresses challenges in unsupervised representation learning, particu-
012 larly in high-fidelity image generation and domain adaptability across diverse
013 data modalities. Current frameworks such as GANs and VQ-VAE have shown
014 promise but face limitations in maintaining consistent performance across vari-
015 able data distributions without significant supervision. To overcome these chal-
016 lenges, we propose a Multi-Modal Vector Quantized Variational AutoEncoder
017 (VQ-VAE) integrated with Self-Supervised Learning (SSL). Our innovative ap-
018 proach incorporates a harmonizer module within the VQ-VAE architecture, which
019 aligns and transforms data representations across multiple modalities. By leverag-
020 ing self-supervised learning techniques, the model iteratively refines its param-
021 eters, enhancing both image reconstruction quality and adaptability to new domains
022 with minimal supervision. The proposed framework processes CIFAR-10 datasets
023 to facilitate structured data integration, employing advanced standardization and
024 batching techniques for optimal performance. Empirical evaluations reveal sub-
025 stantial improvements in image reconstruction fidelity and domain adaptability
026 compared to standard VQ-VAE models, corroborated by metrics such as PSNR,
027 SSIM, and FID. The seamless integration of modality-specific feature extraction
028 and embedding generalization within our framework demonstrates the potential
029 to advance unsupervised learning paradigms. Our contribution establishes a ro-
030 bust solution, optimizing the generative process, and expanding applicability in
031 real-world scenarios characterized by unlabeled, multi-modal datasets.

032 1 INTRODUCTION
033

034 **Research Background:** Unsupervised learning is a pivotal area in artificial intelligence, focusing
035 on identifying significant patterns within large datasets lacking labels. A central challenge in this
036 domain is representation learning (Bengio et al., 2013; LeCun et al., 2015), which involves deriving
037 meaningful features from raw data inputs. Notable methods such as generative adversarial net-
038 works (GANs) (Goodfellow et al., 2014) and vector quantized variational autoencoders (VQ-VAE)
039 (Van Gerven & Bohte, 2017) have achieved substantial success in tasks like high-fidelity image
040 generation and domain adaptation. However, these models often exhibit limitations in maintaining
041 consistency across different data distributions and commonly necessitate extensive supervision with
042 labeled data.

043 Achieving disentangled representations is a critical goal in representation learning, where distinct
044 data characteristics are isolated in separate dimensions. These representations are essential for tasks
045 such as image generation and transfer learning, where understanding data structures without spec-
046 ific supervised tasks is crucial. Despite advances, current VQ-VAE implementations encounter
047 difficulties in handling multi-modal data and capitalizing on unlabeled information to enhance gen-
048 eralization and flexibility.

049 **Research Motivation:** Although current methodologies demonstrate considerable strength, they
050 often fail to maintain reliable performance across varied modalities without significant supervision,
051 limiting their application in diverse real-world contexts. There exists a gap in methods that can inde-
052 pendently fine-tune their parameters to adapt across different modalities without relying heavily on



Meta Review Status: The meta review and decision have not been released yet.

AI Review from DeepReviewer



AI Review from DeepReviewer will be automatically processed



Summary

This paper introduces a Multi-Modal Vector Quantized Variational AutoEncoder (VQ-VAE) integrated with self-supervised learning (SSL) techniques, aiming to enhance image generation and domain adaptation in unsupervised learning settings. The core innovation lies in the incorporation of a 'harmonizer' module within the VQ-VAE architecture, designed to align and transform data representations across different modalities. The authors propose that this harmonizer facilitates coherent translations between modalities by mapping quantized embeddings into a shared latent space. The model is trained using a combination of reconstruction loss and self-supervised learning objectives, which are intended to iteratively refine the model parameters and optimize the embedding space with minimal reliance on labeled data. The authors evaluate their proposed method on the CIFAR-10 dataset, comparing its performance against a baseline VQ-VAE model. The evaluation focuses on image reconstruction fidelity, measured using PSNR, SSIM, and FID scores. The results indicate that the proposed multi-modal VQ-VAE achieves improvements in these metrics compared to the baseline. The paper claims that the proposed method enhances image reconstruction fidelity and domain adaptability, with the harmonizer module playing a key role in aligning representations across different data modalities. However, the experimental validation is limited to the CIFAR-10 dataset, and the claims of domain adaptability and multimodal learning are not fully supported by the experimental design. The paper's contributions are primarily methodological, introducing a novel architecture for VQ-VAE with a harmonizer module and self-supervised learning techniques. While the paper presents a promising approach to multi-modal representation learning, the limited experimental validation and lack of comparison with state-of-the-art methods raise concerns about the generalizability and practical impact of the proposed method. The paper's claims of domain adaptability and multimodal learning are not fully substantiated by the experimental evidence, which is a significant limitation. The paper also lacks a detailed analysis of the computational complexity of the proposed method, which is important for assessing its practicality in real-world applications. Despite these limitations, the paper introduces an interesting approach to multi-modal representation learning that warrants further investigation.



Strengths

The paper presents a well-structured and clearly written description of a novel Multi-Modal VQ-VAE architecture, which is the primary strength of this work. The integration of a harmonizer module within the VQ-VAE framework is a notable technical

innovation, aimed at aligning representations across different data modalities. This approach is clearly articulated, and the authors provide a detailed explanation of the model components, including the encoder, quantizer, harmonizer, and decoder modules. The inclusion of mathematical formulations for key processes, such as the quantization mechanism and reconstruction loss function, enhances the clarity and reproducibility of the proposed method. The authors also clearly describe the use of self-supervised learning techniques to iteratively refine model parameters, which is a valuable aspect of the proposed approach. The paper's methodology section provides a comprehensive overview of the proposed method, making it easy to follow the authors' line of reasoning. The use of a baseline VQ-VAE model for comparison is also a positive aspect of the experimental design, allowing for a direct assessment of the impact of the proposed modifications. The paper's focus on self-supervised learning is also a strength, as it addresses the challenge of learning from unlabeled data, which is a common problem in many real-world applications. The authors' attempt to integrate self-supervised learning with a multi-modal VQ-VAE is a promising direction for future research. The paper's organization is logical, with a clear introduction, methodology, experiments, and results sections, which makes it easy for the reader to follow the authors' line of reasoning. The use of mathematical formulations and diagrams further enhances the clarity of the paper. The paper's focus on a novel architecture for VQ-VAE with a harmonizer module is a valuable contribution to the field of unsupervised representation learning. The authors' attempt to address the challenge of multi-modal representation learning with a novel approach is a promising direction for future research. The paper's clear and concise writing style makes it accessible to a wide audience of researchers and practitioners.

Weaknesses

After a thorough examination of the paper, several significant weaknesses have been identified, which substantially impact the validity and generalizability of the presented findings. Firstly, the paper's evaluation is severely limited by its exclusive reliance on the CIFAR-10 dataset. As the reviewer correctly pointed out, CIFAR-10 is a relatively simple and outdated dataset, which is insufficient to demonstrate the model's capabilities on more complex, real-world scenarios. The paper states, 'The CIFAR-10 dataset, consisting of 60,000 32x32 color images in 10 classes, was used.' This choice of dataset limits the assessment of the model's effectiveness and generalizability, particularly for tasks like image generation and domain adaptation, which require more diverse and challenging datasets. The absence of experiments on more complex and widely used datasets such as ImageNet or COCO is a major shortcoming. This limitation is not just a matter of scale but also of the diversity of visual content, which is crucial for evaluating the robustness of a generative model. The paper's claim that the model achieves 'improved performance in image reconstruction and domain adaptability' is not adequately supported by the experimental results on CIFAR-10 alone. The confidence level for this weakness is high, as the paper explicitly states the use of CIFAR-10 and lacks any evidence of experiments on more complex datasets. Secondly, the paper's comparison with a baseline VQ-VAE model is insufficient to contextualize the performance of the proposed method. The paper states, 'The baseline VQ-VAE model, which does not include the harmonizer module or self-supervised learning techniques, was used for comparison.' While this comparison is useful for understanding the impact of the proposed modifications, it does not provide a clear picture of how the model performs relative to other state-of-the-art models in unsupervised representation learning and image generation. The absence of comparisons with recent methods, such as diffusion-based models, is a significant oversight. These models have demonstrated superior performance in image generation tasks, and their inclusion as baselines would have provided a more comprehensive evaluation of the proposed method's performance. The lack of such comparisons makes it difficult to assess the true impact and novelty of the proposed approach. The confidence level for this weakness is high, as the paper explicitly states the use of a baseline VQ-VAE and lacks any comparisons with other state-of-the-art models. Thirdly, the paper's claim of domain adaptability is not supported by sufficient experimental evidence. The paper states that the model achieves 'domain adaptability,' but there are no experiments that explicitly test the

model's ability to adapt to new, unseen domains. The experiments are limited to the CIFAR-10 dataset, which does not provide any evidence of cross-domain performance. The absence of experiments involving training on one dataset and evaluating on another is a major limitation. This lack of experimental evidence undermines the paper's claim of domain adaptability. The confidence level for this weakness is high, as the paper lacks any cross-domain experiments. Fourthly, the paper's claim of contributing to multimodal learning is not substantiated by the experimental setup. The paper describes a 'Multi-Modal Vector Quantized Variational AutoEncoder (VQ-VAE)' and a 'harmonizer module' designed to align representations across multiple modalities. However, the experiments are limited to the CIFAR-10 dataset, which consists of single-modality image data. The paper states, 'The CIFAR-10 dataset, consisting of 60,000 32x32 color images in 10 classes, was used.' The use of a single-modality dataset does not support the claim of multimodal learning. The absence of experiments with additional modalities, such as text or audio, is a significant shortcoming. This discrepancy between the claim of multimodal learning and the experimental setup is a major weakness. The confidence level for this weakness is high, as the paper explicitly states the use of CIFAR-10 and lacks any experiments with other modalities. Fifthly, the paper lacks a detailed analysis or visualization of the learned embeddings. While the paper discusses the harmonizer module's role in aligning embeddings across modalities, there is no analysis or visualization of the learned embeddings to support this claim. The paper states, 'The Harmonizer aligns quantized embeddings Q within a shared latent space, facilitating coherent translations among data modalities.' However, there are no visualizations, such as t-SNE or PCA plots, to demonstrate the effectiveness of the harmonizer module. The absence of such analysis makes it difficult to verify the harmonizer's effectiveness. The confidence level for this weakness is high, as the paper lacks any embedding visualizations or analysis. Finally, the paper does not provide an analysis of the computational complexity or scalability of the proposed method. The paper describes the architecture, including the harmonizer and SSL module, but there is no discussion of the computational cost associated with the proposed method. The absence of information on training time, memory usage, or computational complexity is a significant oversight. This lack of analysis makes it difficult to assess the practicality of the method for real-world applications. The confidence level for this weakness is high, as the paper lacks any discussion of computational complexity.

Suggestions

To address the identified weaknesses and enhance the overall quality of the paper, several concrete and actionable improvements are recommended. Firstly, the authors should significantly expand their experimental evaluation to include more complex and widely used datasets. Specifically, the inclusion of datasets such as ImageNet or COCO would provide a more rigorous test of the model's capabilities in image generation and domain adaptation. These datasets offer a greater diversity of images and would provide a more realistic assessment of the model's ability to handle real-world scenarios. This change would directly address the limitation of using only the CIFAR-10 dataset. Secondly, the authors should include comparisons with state-of-the-art models in unsupervised representation learning and image generation. This should include recent diffusion-based models, which have demonstrated superior performance in image generation tasks. The inclusion of these models as baselines would provide a more comprehensive understanding of the proposed model's performance relative to existing methods and highlight its unique contributions. This change would directly address the limitation of only comparing against a baseline VQ-VAE. Thirdly, to substantiate the claims of domain adaptability, the authors need to provide explicit experimental evidence demonstrating the model's ability to generalize to new, unseen domains. This could be achieved by training the model on one dataset and evaluating its performance on a different dataset, such as STL-10 or a dataset with different image styles. Such cross-domain experiments would provide a more convincing validation of the model's adaptability claims. This change would directly address the lack of experimental evidence for domain adaptability. Fourthly, to support the claim of multimodal learning, the authors should include experiments with

additional modalities, such as text or audio. This would demonstrate the model's ability to handle multimodal data and provide a more comprehensive evaluation of its capabilities. For example, the authors could explore image-text pairs or image-audio pairs to showcase the model's performance in a truly multimodal setting. This change would directly address the limitation of only using image data. Fifthly, the authors should include a detailed analysis and visualization of the learned embeddings. This should include visualizations, such as t-SNE or PCA plots, of the aligned embeddings across different modalities. This would provide deeper insights into how the harmonizer module functions and its effectiveness in achieving cross-modal alignment. This change would directly address the lack of embedding analysis. Finally, the authors should include an analysis of the model's computational requirements, including training time and memory usage. This is particularly important given the integration of self-supervised learning and the harmonizer module, which may introduce additional computational overhead. An analysis of the model's computational complexity and scalability would help assess its practicality for real-world applications. This change would directly address the lack of computational complexity analysis. These suggestions are all concrete and actionable, and they directly address the identified weaknesses. Implementing these changes would significantly improve the quality and impact of the paper.

? Questions

Several key uncertainties and methodological choices warrant further clarification. Firstly, could the authors provide more details on the architecture of the harmonizer module, such as the number of layers, activation functions, and any specific design choices made? How does the architecture differ when handling different modalities? This question seeks to understand the specific implementation details of the harmonizer module, which is a core component of the proposed method. Secondly, how does the computational complexity of the proposed method compare to the baseline VQ-VAE and other state-of-the-art models in unsupervised representation learning and image generation? Can you provide an analysis of the training time and memory usage of the proposed method? This question seeks to understand the computational cost of the proposed method, which is crucial for assessing its practicality. Thirdly, what specific self-supervised learning techniques are used in your framework? How are the self-supervised objectives integrated into the training process, and how do they contribute to the model's performance? This question seeks to understand the specific details of the self-supervised learning component, which is another key aspect of the proposed method. Fourthly, how does the model handle potential conflicts or inconsistencies between different modalities during the alignment process? Are there any specific strategies or mechanisms in place to ensure coherent and consistent representations across modalities? This question seeks to understand how the model addresses the challenges of aligning representations from different modalities. Fifthly, can you provide more details on the datasets used in the experiments, including any preprocessing steps or data augmentation techniques applied? How do these choices impact the model's performance and generalizability? This question seeks to understand the specific details of the datasets used in the experiments, which is crucial for interpreting the results. Finally, could you provide more information on the hyperparameter settings used in the experiments, such as learning rates, batch sizes, and the number of training epochs? How sensitive is the model's performance to these hyperparameters, and what strategies were used to tune them? This question seeks to understand the specific details of the experimental setup, which is crucial for reproducibility and interpreting the results. These questions target core methodological choices and seek clarification of critical assumptions, which would greatly enhance the understanding of the proposed method.

Scores

Soundness: **1.75**

Presentation: **1.5**

Contribution: **1.5**

Rating: 2.5

AI Review from ZGCA

 AI Review from ZGCA will be automatically processed

Summary

The paper proposes a Multi-Modal VQ-VAE augmented with a novel "harmonizer" module intended to align quantized embeddings across modalities within a shared latent space, and integrates self-supervised learning (SSL) to improve representation quality and domain adaptability. The architecture consists of an encoder, a VQ quantizer (codebook size 512), the harmonizer (linear transforms + layer norm), and a decoder. The method is evaluated on CIFAR-10 using PSNR, SSIM, and FID, and reports improvements over a baseline VQ-VAE (e.g., PSNR 29.76 vs 27.43). The paper claims comprehensive ablations and statistical significance testing, and argues that the approach enhances multi-modal learning and domain adaptability.

Strengths

- Clear motivation to improve unsupervised/multi-modal representation learning and image reconstruction with minimal supervision (Introduction; Abstract).
- Conceptually interesting idea of an explicit harmonization layer inside a VQ-VAE to align quantized embeddings across modalities (Sections 3.2.3, 4.2.3).
- Simple, modular architecture description for encoder/quantizer/decoder that is easy to follow at a high level (Section 4.2).

- Reported quantitative improvements on CIFAR-10 over a baseline VQ-VAE in PSNR/SSIM/FID (Table 1; Section 4.4).

✗ Weaknesses

- Core claim (multi-modal alignment and translation via the harmonizer) is not empirically validated; all experiments are single-modal on CIFAR-10 (Abstract; Sections 3.1, 4.4).
- Harmonizer mechanism is under-specified (only noted as linear layers + layer norm) with no concrete formulation of its training objective for cross-modal alignment, no architectural diagram, and no analysis of its behavior (Sections 3.2.3, 4.2.3).
- SSL is described inconsistently: while the text references contrastive learning (SimCLR) and BYOL (Sections 2.3, 3.3, 4.3), the only explicitly defined loss is L2 reconstruction (Eq. for L_r), which is standard reconstruction training rather than a self-supervised objective. No contrastive/auxiliary task, projection heads, or positive/negative construction is specified.
- VQ-VAE training details are incomplete: standard codebook and commitment losses (and straight-through estimator) are not described, and the overall objective is unclear. Hyperparameters, optimizer, training schedule, data augmentations, latent dimension d , and other crucial settings are missing (Sections 3.2.2–3.3, 4.2).
- Baseline VQ-VAE is not specified (architecture, training, losses, hyperparameters), making the comparison potentially unfair and not reproducible (Section 4.4).
- Claims of "comprehensive ablation studies" and "statistical significance tests" are not accompanied by any presented ablation tables/figures or statistical details (Section 4.4).
- Domain adaptability and multi-modal capabilities are asserted but not tested; there are no domain shift experiments (e.g., CIFAR-10-C, CIFAR-10 $\rightarrow$ STL-10) or evaluations on multi-modal datasets (Sections 1, 4.4).
- FID is reported but the sampling procedure is unspecified; it appears to be computed on reconstructions rather than samples from a learned prior, which limits interpretability as a generative metric (Section 4.4).
- Writing is repetitive and some sections are truncated or vague (e.g., Section 4.4 ends mid-sentence), and several references are generic or slightly mismatched to the specific method choices.

? Questions

- Multi-modal validation: What modalities does the harmonizer target? Can you provide experiments on a truly multi-modal dataset (e.g., image–text like MS-COCO, audio–visual, or multi-sensor) demonstrating cross-modal alignment, transfer, and robustness with partial modalities?
- Harmonizer specifics: Please provide a precise mathematical description of the harmonizer, its input/output tensors, parameterization, and training objective. How does it enforce alignment across modalities? Is there a contrastive or matching loss between modality-specific codes?

- VQ-VAE objective: What exact losses are used? Do you include the standard codebook and commitment losses with a straight-through estimator (as in van den Oord et al., 2017)? What are the coefficients? How is the codebook updated?
- SSL objective and pipeline: If contrastive learning is employed, please specify augmentations, projection heads, positive/negative construction, temperature, batch size, and how the contrastive loss is combined with reconstruction/VQ losses. If using BYOL-style objectives, please detail the online/target networks and momentum updates.
- Baseline details: What is the baseline VQ-VAE architecture, training schedule, hyperparameters, codebook size, and loss configuration? Did you re-implement or use a standard implementation? How did you ensure a fair comparison?
- Ablations: Please provide the promised ablation results for: (a) removing the harmonizer; (b) removing SSL; (c) varying codebook size; (d) varying latent dimensionality; (e) different harmonizer designs. Include mean $\pm$ std over multiple runs.
- Statistical testing: How many runs were conducted for the reported metrics? Please report mean, standard deviation, and statistical tests (e.g., p-values or confidence intervals).
- Domain adaptability: Can you include experiments under domain shift (e.g., CIFAR-10-C corruptions or CIFAR-10 $\rightarrow$ STL-10) to substantiate adaptability claims?
- FID computation: Are FID scores computed on reconstructions or on samples generated from a learned prior over latents? If the latter, please describe the prior and sampling procedure. If the former, please clarify and consider reporting LPIPS/reconstruction FID explicitly.
- Reproducibility: Please provide full training details (optimizer, learning rate schedule, batch size, epochs, data augmentations, early stopping), latent dimensionality d , model parameter counts, wall-clock time/compute, and code availability plans.
- Qualitative analysis: Can you include qualitative reconstructions/samples, codebook usage statistics, and visualization of latent alignment (e.g., t-SNE/UMAP across modalities) to demonstrate the harmonizer's effect?

Limitations

- Evaluation is restricted to a single small-scale dataset (CIFAR-10) with no multi-modal or domain-shift experiments, limiting external validity.
- Methodological under-specification (harmonizer mechanism, SSL objective, VQ-VAE losses) hinders reproducibility and interpretability.
- FID reported without clear sampling protocol; likely computed on reconstructions, which limits conclusions about generative modeling quality.
- Potential compute cost and codebook efficiency trade-offs are not analyzed; scalability to higher-resolution or multiple modalities is untested.
- Potential negative societal impacts: as with many generative models, risks include misuse for synthetic content generation and amplification of dataset biases; no mitigation strategies are discussed.

Cross-Modal Consistency: 28/50

Textual Logical Soundness: 14/30

Visual Aesthetics & Clarity: 16/20

Overall Score: 58/100

Detailed Evaluation (≤ 500 words):

1. Cross-Modal Consistency

- Major 1: Claims of multimodality and domain adaptability are not supported by visuals/experiments; only CIFAR-10 single-domain results are shown. Evidence: Sec 4.4, Table 1; Abstract; Contributions.
- Major 2: SSL is described as contrastive, yet only MSE reconstruction loss is specified; no contrastive objective/augments are shown. Evidence: Sec 4.3; equation for L_r .
- Minor 1: “Comprehensive ablation studies” are claimed but absent; only a single table is provided.
- Minor 2: Quantization equation rendered as “Q u a n t i z e(·)” and no codebook training details (EMA/commitment loss) shown.
- Minor 3: Table 1 supports “statistical significance tests” claim in text, but no test statistics or variances are presented.

2. Text Logic

- Major 1: Flagship claims (multi-modal alignment, domain adaptability with minimal supervision) lack a demonstrated evaluation protocol or datasets beyond CIFAR-10. Evidence: Sec 4.4, Table 1; Abstract.
- Major 2: Mis-citation of VQ-VAE origin in Intro undermines foundational grounding. Evidence: “vector quantized variational autoencoders (VQ-VAE) (Van Gerven & Bohte, 2017)”.
- Minor 1: Preprocessing ambiguously mixes 0–1 scaling and zero-mean/unit-variance without a clear chosen setting or per-channel details.

3. Figure Quality

- No Major issues found.
- Minor 1: Table 1 is readable but lacks standard deviations, sample size, or protocol (reconstructions vs. samples) to judge FID/PSNR/SSIM reliability.
- Minor 2: No qualitative images (reconstructions/samples) are provided; the reader cannot visually verify fidelity or artifacts.
- Minor 3: “Figure-Alone” test: Table conveys basic comparison but not evaluation protocol; add caption details (dataset split, compute settings).

Key strengths:

- Clear modular architecture description (encoder, quantizer with 512 entries, harmonizer, decoder).
- Simple, readable presentation; Table 1 indicates non-trivial PSNR/SSIM/FID gains over a baseline.

Key weaknesses:

- Central claims (multimodality, domain adaptation, SSL contrastive gains) are unsupported by experiments.
- SSL formulation lacks contrastive objective/augmentations; current loss is purely reconstruction.
- Missing ablations/qualitative results and statistical reporting; mis-citation of VQ-VAE origin.

Recommendations:

- Provide multi-domain/multi-modality experiments and a domain-shift protocol; add qualitative visuals.
- Implement and report a true SSL/contrastive loss (augmentations, temperature, positives/negatives) and ablate harmonizer/SSL/codebook.
- Correct citations and detail training specifics (commitment loss, EMA, FID protocol, seeds, std/error bars).

 Scores

Originality: 2

Quality: 1

Clarity: 2

Significance: 1

Soundness: 1

Presentation: 2

Contribution: 2

Rating: 3

AI Review from SafeReviewer

 AI Review from SafeReviewer will be automatically processed

 Summary

This paper introduces a Multi-Modal Vector Quantized Variational AutoEncoder (VQ-VAE) integrated with Self-Supervised Learning (SSL) techniques, aiming to enhance image generation and domain adaptability. The core idea revolves around a novel 'harmonizer' module designed to align representations across different modalities within the VQ-VAE framework. The authors propose that this harmonizer, combined with self-supervised learning, will improve image reconstruction quality and adaptability to new domains with minimal supervision. The methodology involves a standard VQ-VAE architecture with the addition of the harmonizer, which is described as employing linear transformations followed by layer normalization to align quantized embeddings in a shared latent space. The self-supervised learning component is mentioned as leveraging contrastive learning frameworks to minimize reconstruction loss, but the specific details of this integration are not fully elaborated. The experimental evaluation is primarily conducted on the CIFAR-10 dataset, where the proposed method is compared against a baseline VQ-VAE model. The results, presented in terms of PSNR, SSIM, and FID scores, suggest that the proposed approach achieves some improvements over the baseline. However, the paper lacks a comprehensive analysis of the harmonizer's impact, and the experimental scope is limited, raising questions about the generalizability of the findings. Overall, while the paper presents an interesting concept, the lack of detailed explanations, limited experimental validation, and several presentation issues significantly weaken its impact and credibility.

Strengths

While the paper presents a novel concept, the strengths are somewhat limited due to the issues identified in the weaknesses. However, I can acknowledge the attempt to integrate a harmonizer module within the VQ-VAE framework as a potential contribution. The idea of aligning representations across multiple modalities is relevant, especially in the context of improving generalization and adaptability. The use of self-supervised learning to refine model parameters is also a promising direction. The paper's structure, while needing improvement, does follow a standard academic format, which suggests an understanding of the expected presentation style. The inclusion of a 'Related Work' section, despite its lack of in-text citations, indicates an attempt to contextualize the research within the existing literature. The authors' choice of PSNR, SSIM, and FID as evaluation metrics is appropriate for assessing image reconstruction quality, although the lack of detailed analysis of these results is a weakness. The paper also attempts to address the challenge of domain adaptability, which is a significant problem in machine learning. Despite the limitations, the core idea of using a harmonizer to align multi-modal representations within a VQ-VAE framework is a potentially valuable contribution to the field.

Weaknesses

My analysis reveals several significant weaknesses that undermine the paper's credibility and impact. First, the paper suffers from a lack of clarity and precision in its writing. There are instances of spelling errors, such as 'paterns' instead of 'patterns' in the Introduction, and 'paradigms' instead of 'parameters' in the Method section. These errors, while seemingly minor, detract from the overall professionalism of the work. More importantly, the paper lacks in-text citations, which is a

major oversight. For example, the 'Related Work' section discusses previous studies by Agarwal et al., Krizhevsky et al., and others without ever citing them in the main text. This makes it difficult to assess the authors' understanding of the existing literature and their contribution relative to it. The lack of in-text citations is a serious issue that affects the paper's credibility.

Second, the paper's claims are not always well-supported by evidence. The authors claim to be using a 'Multi-Modal' VQ-VAE, but the experiments are conducted solely on the CIFAR-10 dataset, which consists of single-modality images. There is no evidence presented to demonstrate the model's ability to handle multiple modalities. This discrepancy between the claim and the experimental setup is a major weakness. Furthermore, the paper lacks a detailed explanation of the 'harmonizer' module. While the authors describe it as employing linear transformations followed by layer normalization, the specific mathematical operations and the rationale behind this design choice are not clearly articulated. This lack of detail makes it difficult to understand the harmonizer's function and its contribution to the overall performance. The paper also claims that the harmonizer facilitates 'coherent translations among data modalities' and ensures 'smooth latent vector transitions,' but these claims are not supported by any experimental evidence or analysis. The lack of a detailed explanation of the harmonizer is a significant weakness that hinders the understanding of the proposed method.

Third, the experimental evaluation is limited and lacks rigor. The experiments are conducted solely on the CIFAR-10 dataset, which is relatively simple compared to more complex datasets like ImageNet. The paper compares the proposed method against a 'Baseline VQ-VAE' model, but there is no information about the implementation details of this baseline, making it difficult to assess the fairness of the comparison. Furthermore, the paper does not compare the proposed method against other relevant baselines, such as standard VQ-VAE or VQ-VAE-2, or other multi-modal representation learning methods. The lack of a comprehensive comparison with existing methods is a major weakness that limits the assessment of the proposed method's effectiveness. The paper also lacks a detailed analysis of the experimental results. While the authors present PSNR, SSIM, and FID scores, there is no in-depth discussion of these results or any visualizations of the learned representations or the harmonizer's effect. The absence of a detailed analysis of the experimental results is a significant weakness that limits the understanding of the proposed method's performance. Finally, the paper's overall presentation is problematic. The title is too broad and does not accurately reflect the content of the paper. The methodology section is split into two parts, which is confusing and deviates from standard academic writing practices. The paper also lacks a clear explanation of how the self-supervised learning component is integrated with the VQ-VAE framework. The lack of clarity in the presentation and the missing details about the self-supervised learning integration are significant weaknesses that hinder the understanding of the proposed method. The combination of these issues significantly undermines the paper's credibility and impact.

Suggestions

Based on the identified weaknesses, I recommend several concrete improvements. First and foremost, the authors must thoroughly revise the paper to include proper in-text citations. This is a fundamental requirement for academic writing and is essential for establishing the paper's credibility. The authors should carefully review the 'Related Work' section and ensure that all mentioned studies are properly cited in the main text. Second, the authors need to provide a more detailed explanation of the 'harmonizer' module. This should include the specific mathematical operations involved, the rationale behind the design choices, and a clear explanation of how it achieves alignment across modalities. The authors should also provide a more detailed explanation of how the self-supervised learning component is integrated with the VQ-VAE framework. This should include the specific loss functions used and the optimization process. Third, the authors should

expand the experimental evaluation to include more complex datasets and more relevant baselines. The use of CIFAR-10 alone is insufficient to demonstrate the effectiveness of the proposed method, especially given the claim of multi-modal learning. The authors should consider using datasets that include multiple modalities, such as image-text datasets. The authors should also compare their method against standard VQ-VAE, VQ-VAE-2, and other multi-modal representation learning methods. This will provide a more comprehensive assessment of the proposed method's performance. Fourth, the authors should provide a more detailed analysis of the experimental results. This should include visualizations of the learned representations, an analysis of the harmonizer's effect, and a discussion of the limitations of the proposed method. The authors should also provide a more detailed explanation of the baseline implementation to ensure a fair comparison. Fifth, the authors should revise the paper's presentation to improve its clarity and readability. This should include splitting the 'Methodology' section into 'Preliminaries' and 'Proposed Method,' providing a more specific title, and ensuring that the claims made in the paper are supported by evidence. Finally, the authors should carefully proofread the paper to correct any spelling errors or grammatical mistakes. These changes are essential for improving the paper's credibility and impact.

Questions

Several key uncertainties remain after my analysis, and I would like to pose the following questions to the authors. First, regarding the 'harmonizer' module, could you provide a more detailed explanation of the specific mathematical operations involved? How does the linear transformation followed by layer normalization achieve alignment across different modalities? What is the rationale behind this specific design choice? Second, concerning the self-supervised learning component, could you provide more details on how it is integrated with the VQ-VAE framework? What specific loss functions are used for self-supervised learning, and how do they interact with the VQ-VAE loss? How does the self-supervised learning component contribute to the overall performance? Third, regarding the experimental evaluation, why was the CIFAR-10 dataset chosen as the primary evaluation dataset, given the claim of multi-modal learning? What other datasets were considered, and why were they not used? Could you provide more details on the implementation of the baseline VQ-VAE model? How was it trained, and what hyperparameters were used? Fourth, regarding the claims of domain adaptability, could you provide any experimental evidence to support this claim? What experiments were conducted to demonstrate the model's ability to adapt to new domains? What metrics were used to evaluate domain adaptability? Finally, regarding the overall presentation, why was the methodology section split into two parts? What was the reasoning behind this decision? Could you provide a more specific title that accurately reflects the content of the paper? Addressing these questions will help clarify the paper's contributions and limitations.

Scores

Soundness: **1.0**

Presentation: **1.0**

Contribution: **1.0**

Rating: 1.0

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